

Insurance Dataset Attributes

A simple explanation of what each variable means and how to understand the dataset.

1. Big Picture

One row can be understood as one insured person in one calendar year. The attributes describe who the person is, which policy they have, how long they were covered, how much they paid, how much healthcare they used, whether they left the insurer, and characteristics of their location.

2. Identification Attributes

Attribute	Simple explanation
ID	Unique record identifier, formed from the policy and insured-person identifiers.
ID_policy	Identifies the insurance contract. Several people can share the same policy.
ID_insured	Identifies the individual within a policy. Example: 1 = father, 2 = mother, 3 = child.

3. Time and Date Attributes

Attribute	Simple explanation
period	Calendar year being represented, such as 2024.
date_effect_insured	Date the individual started being covered by the insurer.
date_lapse_insured	Date the individual's coverage ended. Empty usually means still covered.
date_effect_policy	Date the insurance contract itself started.
date_lapse_policy	Date the insurance contract itself ended.
year_effect_insured	Year in which the individual became insured.
year_lapse_insured	Year in which the individual's insurance ended.
year_effect_policy	Year in which the policy contract started.
year_lapse_policy	Year in which the policy contract ended.

4. Exposure and Lapse

Attribute	Simple explanation
exposure_time	Fraction of the year the person was actually covered. 1 = full year; 0.5 ≈ half year; 0 = no effective coverage.
lapse	Status of the insured person during the year: 1 = left before year-end, 2 = active, 3 = ended at year-end.

5. Seniority

Attribute	Simple explanation
seniority_insured	How many years the individual has been associated with the insurer.
seniority_policy	How many years the policy contract has existed with the insurer.

6. Insurance Product

Attribute	Simple explanation
type_policy	Basic policy type: I = Individual, C = Collective.
type_policy_dg	More detailed type: S = Self-employed, I = Individual, C1-C4 = different collective categories.
type_product	Coverage type: D = Dental, P = Premium, S = Standard, I = International.
reimbursement	Whether the product includes reimbursement of medical expenses: Yes/No.
new_business	Whether the contract is new business (1) or a renewal/existing contract (0).
distribution_channel	How the policy was sold: A = Agency, D = Direct, I = Insurance intermediary.

7. Customer Demographics

Attribute	Simple explanation
gender	Gender of the insured: M = Male, F = Female.
age	Age of the insured in whole years, rounded from the exact age.

8. Financial Attributes

Attribute	Simple explanation
premium	Annual net premium associated with the insured person, in euros.
cost_claims_year	Total healthcare claim cost generated by the insured person during the year, in euros.

9. Healthcare Usage

Attribute	Simple explanation
n_medical_services	Number of medical services used during the year, such as consultations, tests, or procedures.

10. Geographic / Population Attributes

Attribute	Simple explanation
n_insured_pc	Number of the insurer's insured people in the same postal-code area.
n_insured_mun	Number of the insurer's insured people in the same municipality.
n_insured_prov	Number of the insurer's insured people in the same province.
IICIMUN	Number of the insurer's insured people per 1,000 inhabitants at municipality level.
IICIPROV	Number of the insurer's insured people per 1,000 inhabitants at province level.
C_H	Municipality population category: H1 ≤1,000; H2 1,001-10,000; H3 10,001-25,000; H4 25,001-50,000; H5 50,001-100,000; H6 >100,000.

11. Income Attributes

Attribute	Simple explanation
C_GI	Income percentile of the person's postal code compared with the general/national population.
C_II	Income percentile of the person's postal code compared with the insurer's insured population.

12. Education Attributes

Attribute	Simple explanation
C_IE_P	Postal-code education category/percentile for primary education among the insured population.
C_IE_S	Postal-code education category/percentile for secondary education among the insured population.
C_IE_T	Postal-code education category/percentile for tertiary/higher education among the insured population.
C_GE_P	Postal-code education category/percentile for primary education among the general population.
C_GE_S	Postal-code education category/percentile for secondary education among the general population.
C_GE_T	Postal-code education category/percentile for tertiary/higher education among the general population.

13. Climate / Regional Attribute

Attribute	Simple explanation
C_C	Climate/geographic cluster: C1 = Central-continental, C2 = Canary Islands, C3 = Mediterranean, C4 = North-continental, C5 = South-continental, C6 = Oceanic.

14. How to Think About the Dataset

WHO? ID, ID_policy, ID_insured, age, gender

WHEN? period, effect/lapse dates, effect/lapse years

WHAT INSURANCE? policy type, product, reimbursement, channel, new business

HOW LONG? exposure_time, seniority_insured, seniority_policy

DID THEY LEAVE? lapse

MONEY? premium, cost_claims_year

HEALTHCARE USE? n_medical_services

WHERE? postal code/municipality/province counts, income, education, population, climate

15. Example of One Row

Imagine a row contains: age = 35, gender = F, type_policy = C, type_product = P, premium = €1,500, cost_claims_year = €800, n_medical_services = 6, exposure_time = 1.0, lapse = 2, seniority_insured = 5, and seniority_policy = 10.

In simple words: **A 35-year-old woman has a collective premium insurance policy. She was covered for the full year, paid about €1,500, generated €800 in claims, used 6 medical services, has been insured for 5 years, the policy has existed for 10 years, and she remained active at the end of the year.**

16. Useful Classification for Data Analysis

Type	Examples
Identifiers	ID, ID_policy, ID_insured
Dates	date_effect_insured, date_lapse_insured, date_effect_policy, date_lapse_policy
Categorical	gender, type_policy, type_product, reimbursement, new_business, distribution_channel, lapse, C_H, C_C
Numerical	age, exposure_time, seniority_insured, seniority_policy, premium, cost_claims_year, n_medical_services, IICI* and population counts
Year fields	year_effect_insured, year_lapse_insured, year_effect_policy, year_lapse_policy

Important for machine learning: ID, ID_policy and ID_insured are identifiers, not ordinary predictive features. They are mainly useful for tracking individuals/policies, checking continuity across years, and creating appropriate train/test splits.